TRANSPORT AND TELECOMMUNICATION INSTITUTE

Laboratory Work Lab# 1/Part2 /TASK

Topic: Comparison of AI Tools for Writing Program Code

Objective

To study the capabilities of modern AI tools for generating program code, compare their strengths and weaknesses, and evaluate how useful they are for a beginner programmer.

Theoretical Introduction

Artificial intelligence is actively used in programming. Modern AI tools can:

- generate program code;
- explain how a program works;
- detect and fix errors;
- suggest improvements.

However, the results produced by AI are not always fully correct, so a programmer must be able to analyze, verify, and evaluate the generated code.

Task

It is required to compare the performance of three AI tools available via a web browser:

- ChatGPT - https://chat.openai.com
- Claude - https://claude.ai
- Gemini - https://gemini.google.com

Each tool must be given the same task, and then the obtained results should be compared.

Practical Part

Task for All Tools

Ask each AI to write a program in one selected programming language:
(Python / C# / Java)

Program Requirements

Develop a program called **"Number Check"** that:

- asks the user to input a number;
- determines whether the number is:
  - even or odd;
  - positive, negative, or zero;
- displays the result on the screen.

Unified Prompt for All AI Tools

The student must use the same prompt for all three tools.

Example Prompt:

Write a program in Python that asks the user to enter a number, determines whether it is even or odd, positive or negative, and displays the result. Explain the code in simple terms.

The programming language may be changed according to the instructor's instructions.

Procedure

1. Preparation

- Open three AI tools:
  - ChatGPT
  - Claude
  - Gemini
- Choose one programming language.
- Copy the same prompt into all three tools.

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2. Obtaining Results

For each tool, it is necessary to:

- enter the prompt;
- obtain the generated code;
- save the response;

- check whether the program works correctly;
- if necessary, copy the code into a development environment and run it.

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3. Analysis of Results

For each AI tool, evaluate the following:

3.1. Understanding of the Task

- whether the tool correctly understood the task;
- whether all requirements were fulfilled;
- whether additional clarifications were needed.

3.2. Code Quality

- whether the program works correctly;
- whether there are syntax or logical errors;
- whether the code corresponds the task requirements.

3.3. Clarity for the Student

- whether an explanation is provided;
- whether the code is understandable for a beginner;
- whether comments are included.

3.4. Usability

- whether the interface is convenient;
- response speed;
- ease of copying and reading the code.

3.5. Usefulness of the Result

- whether the result can be used for learning;
- whether the tool helps understand program logic;
- whether improvements are suggested.

Additional Part

After receiving the initial response, perform an additional step.

Additional Task

Ask each AI to improve the program by adding handling of incorrect user input.

Example Prompt:

Improve the program so that it handles errors when the user enters a non-numeric value.

After that, compare the responses again.

Report Format

The report must include the following sections:

1. Title Page

- name of the educational institution;
- subject;
- topic of the laboratory work;
- student’s full name;
- group;
- date.

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2. Objective

Briefly state the objective of the laboratory work.

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3. AI Tools Used

List the tools:

- ChatGPT
- Claude
- Gemini

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4. Initial Task

Provide the task given to the AI.

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5. Prompt Used

Provide the same prompt that was sent to all three tools.

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6. Results Obtained

For each tool, indicate:

- name;
- generated code;
- brief description of the result.

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7. Comparison Table

Criterion	ChatGPT	Claude	Gemini
Understanding of the task			
Completeness of the solution			
Code correctness			
Clarity of explanation			
Usability			
Program improvement			

You may use a 5-point grading scale:

- 5 — excellent
- 4 — good
- 3 — satisfactory
- 2 — poor
- 1 — very poor

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8. Analysis of Results

In this section, the student should answer:

- which tool understood the task best;
- which tool provided the highest-quality code;
- which response was the most understandable;
- which tool was the most convenient to use;
- whether AI can be fully trusted in programming.

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9. Conclusion

Formulate an overall conclusion of the laboratory work.

1. What are AI tools for programming?
2. What advantages do AI tools provide in coding?
3. What are the disadvantages of AI-generated code?
4. Why should generated code always be verified?
5. Which of the tested tools was the most useful for a beginner programmer?
6. Would you trust AI to write real code?